

CAMAGUEY INTL, Cuba

WMO: 782550

Lat: 21.420N

Lon: 77.848W

Elev: 126

StdP: 99.82

Time Zone: -5.00 (CUB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	16.6	9.3	7.4	20.4	10.9	8.2	21.2	11.1	25.1	9.7	25.9	2.3	10	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	33.8	23.6	33.1	23.8	32.6	23.8	26.2	29.9	25.8	29.5	25.5	29.3	3.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	27.2	25.0	20.4	27.1	24.3	19.5	26.8	81.9	30.0	80.1	29.9	78.4	29.3	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.4	7.5	12.8	35.0	1.5	0.8	11.7	35.5	10.8	36.0	10.0	36.4	8.9	37.0		
			10.8	26.8	1.7	0.7	9.5	27.3	8.6	27.7	7.6	28.1	6.4	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.7	22.9	23.6	24.2	25.6	26.5	27.4	28.0	28.0	27.6	26.6	24.5	23.8
	DBStd	2.33	2.14	1.96	1.95	1.42	1.26	1.08	0.87	0.91	0.91	1.14	1.66	1.89
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	3	2	1	0	0	0	0	0	0	0	0	0	1
	CDD10.0	5738	400	382	441	467	511	521	558	527	514	434	426	
	CDD18.3	2700	143	149	183	217	252	271	299	300	277	256	184	169
	CDH23.3	21240	768	966	1302	1639	2034	2446	2902	2854	2427	1973	1042	888
	CDH26.7	7831	188	291	452	605	803	981	1228	1199	968	671	238	208
Wind	WSAvg	3.7	3.9	4.0	4.1	4.0	3.7	3.5	3.8	3.5	2.9	3.2	3.6	3.8
Precipitation	PrecAvg	1369	32	29	53	90	212	199	141	163	212	162	69	37
	PrecMax	1703	131	112	144	211	425	339	292	308	341	754	282	95
	PrecMin	1070	0	0	0	18	42	85	49	28	112	28	1	2
	PrecStd	145	31	32	39	45	106	68	66	64	55	145	63	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	32.0	33.2	33.2	34.0	34.0	34.2	34.2	33.9	32.7	31.1	31.1
		MCWB	20.7	20.4	20.9	21.6	22.6	24.2	23.6	24.1	24.1	23.9	23.7	22.3
	2%	DB	29.9	30.9	31.9	32.1	32.8	33.0	33.2	33.2	32.9	31.9	30.1	30.0
		MCWB	21.2	20.5	20.8	21.7	22.8	24.3	23.9	24.3	24.3	24.1	23.3	22.5
	5%	DB	28.9	29.9	30.8	31.1	31.9	32.1	32.9	32.9	32.1	31.0	29.1	29.0
		MCWB	21.1	20.8	20.7	21.7	22.7	24.3	24.0	24.3	24.4	24.1	23.1	22.1
	10%	DB	27.8	28.8	29.2	30.1	30.9	31.3	32.1	32.0	31.2	30.2	28.0	27.9
		MCWB	21.0	20.7	20.6	21.7	22.6	24.3	24.3	24.5	24.5	24.2	22.6	22.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.2	23.4	25.0	25.6	26.9	26.3	26.6	26.5	26.2	25.4	24.6
		MCDB	27.0	27.3	27.5	28.8	29.1	30.9	30.0	30.7	30.4	29.2	27.9	27.6
	2%	WB	22.6	22.5	22.7	24.0	24.8	26.1	25.8	26.0	25.9	25.7	24.6	23.7
		MCDB	26.4	26.6	27.0	28.0	28.8	29.4	29.7	30.2	29.8	28.8	27.7	27.0
	5%	WB	22.2	22.0	22.3	23.4	24.4	25.6	25.4	25.6	25.5	25.2	24.1	23.3
		MCDB	25.9	26.1	26.6	27.5	28.3	29.0	29.5	29.8	29.5	28.5	27.2	26.5
	10%	WB	21.6	21.5	21.8	22.8	23.9	25.1	25.0	25.1	25.0	24.8	23.6	22.7
		MCDB	25.3	25.7	26.1	27.0	27.8	28.8	29.4	29.5	29.2	28.2	26.4	25.9
Mean Daily Temperature Range	5% DB	MDBR	9.0	9.9	10.3	9.9	9.4	8.9	9.2	9.2	8.8	7.8	7.4	8.1
		MCDBR	10.3	11.4	11.9	11.0	10.5	9.5	9.8	9.9	9.4	8.7	8.2	9.4
		MCWBR	3.6	3.5	3.4	3.0	3.1	3.4	3.2	3.4	3.3	3.3	3.3	3.4
	5% WB	MCDBR	9.1	9.9	10.2	9.6	9.1	8.9	9.1	9.1	8.6	7.8	7.0	8.2
		MCWBR	3.6	3.5	3.5	3.4	3.4	3.8	3.5	3.6	3.5	3.4	3.3	3.3
Clear-Sky Solar Irradiance	taub	0.370	0.377	0.394	0.417	0.450	0.495	0.491	0.470	0.427	0.411	0.394	0.426	
	taud	2.487	2.466	2.414	2.346	2.293	2.160	2.163	2.244	2.391	2.434	2.435	2.283	
	Ebn at Noon	902	916	911	889	851	806	810	831	868	870	868	822	
	Edh at Noon	100	108	119	129	135	153	152	141	120	111	105	120	
All-Sky Solar Radiation	RadAvg	4.18	5.05	5.73	6.28	6.04	6.02	6.46	6.23	5.63	4.84	4.25	3.93	
	RadStd	0.22	0.25	0.27	0.28	0.39	0.32	0.28	0.19	0.29	0.36	0.26	0.30	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page